

Mir Mahathir Mohammad - Curriculum Vitae

Contact

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Portfolio: https://mirmahathir1.github.io/ LinkedIn GitHub Google Scholar DBLP

Education

University of Utah, Salt Lake City, Utah Ph.D. in Computer Science Kahlert School of Computing August 2023 – Present Advisor: El Kindi Rezig	University of Utah, Salt Lake City, Utah M.S. in Computer Science Kahlert School of Computing August 2023 – December 2025 CGPA: 3.978 / 4.0 Advisor: El Kindi Rezig	Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh B.S. in Computer Science April 2017 – April 2022 Advisor: Muhammad Abdullah Adnan
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Research Interests

I focus on data preparation in general and data discovery in particular for data lakes. I built SemDisc, the first end-to-end semantic join discovery system using a query-by-example interface. I also explored how discovered relationships can recommend meaningful ways to organize and sort data. I built NLDisc, which turns a user's natural language description of intent into join discovery over data lakes, and co-developed LORY-J, which treats spatial attributes as first-class signals to discover location-aware joins. I contributed to Buckaroo, a visual data wrangling system that enables users to interactively clean and repair data anomalies in visualizations. My interests include:
Data Systems • Data Lakes • Data Discovery & Integration • Data Wrangling • Data Cleaning • AI for Data Management

Publications

Publications since starting Ph.D. (2023-Present, ~3.2 years)

- Under Review (SIGMOD'27): Mir Mahathir Mohammad, El Kindi Rezig. "Less is More: Leveraging Small Language Models and Join Discovery in Data Lakes for Missing Value Imputation." a missing value imputation system from joins across multiple data lake tables using a small language model
- SIGMOD'26: Mir Mahathir Mohammad, El Kindi Rezig. "Qualitative Join Discovery in Data Lakes using Examples." Accepted at ACM SIGMOD International Conference on Management of Data (SIGMOD'26), 2026. [PDF] [Link] a system for discovering hybrid join paths (combining semantic and equi-joins) in data lakes using query-by-example, supporting hidden tables and semantic tuple matching
- VLDB'26 (Demo): Lu Xing, Mir Mahathir Mohammad, Walid G. Aref, El Kindi Rezig. "LORY-J: Location-aware Join Discovery from Data Lakes." Accepted at VLDB 2026 (Demo Track). [PDF] [Link] a system for location-aware join discovery in data lakes that treats spatial attributes as first-class signals, composing hybrid join paths through an interactive, map-driven interface
- SIGMOD'26 (Demo): Mir Mahathir Mohammad, El Kindi Rezig. "SemDisc: An End-to-End Query-by-Example Semantic Join Discovery System." Accepted at ACM SIGMOD 2026 (Demo Track). [PDF] [Link] an interactive demo enabling users to discover hybrid join paths in schema-less data lakes through a query-by-example interface
- SIGMOD'26 (Demo): Mir Mahathir Mohammad, El Kindi Rezig. "NLDisc: Intent-aware Join Discovery using Natural Language." Accepted at ACM SIGMOD 2026 (Demo Track). [PDF] [Link] a system for intent-aware join discovery over data lakes using natural language queries with interactive intent disambiguation and join path diversification
- CIDR'26: El Kindi Rezig, Mir Mahathir Mohammad, Nicolas Baret, Ricardo Mayerhofer, Andrew McNutt, Paul Rosen. "Towards Scalable Visual Data Wrangling via Direct Manipulation." Accepted at CIDR 2026. [PDF] [Link] a visual data wrangling system that enables users to clean and repair data anomalies through direct manipulation of interactive visualizations

- **VLDB'25 (Demo):** Akash Khatri, **Mir Mahathir Mohammad**, El Kindi Rezig. "Sort it Like You Mean It: Discovering Semantically Interesting Attribute Augmentations to Sort Tables." Accepted at VLDB 2025 (Demo Track). [\[PDF\]](#) [\[Link\]](#)
Recommends semantically meaningful ways to sort tables by automatically discovering and augmenting attributes from data lakes using LLMs.

Undergraduate research

- **IEEE FG'24:** Iftekhar E Mahbub Zeeon, **Mir Mahathir Mohammad**, Muhammad Abdullah Adnan. "BTVSL: A Novel Sentence-Level Annotated Dataset for Bangla Sign Language Translation." Accepted at IEEE FG 2024. *Introduces the first large-scale sentence-level dataset for Bangla Sign Language translation.* [\[PDF\]](#) [\[Link\]](#)
- **Neurocomputing'22:** Md. Ashraful Islam, **Mir Mahathir Mohammad**, Sarkar Snigdha Sarathi Das, Mohammed Eunus Ali. "A survey on deep learning based Point-of-Interest (POI) recommendations." Accepted at Neurocomputing (Journal), 2022. [\[PDF\]](#) [\[Link\]](#) *Categorizes deep learning approaches for POI recommendation systems in location-based social networks.*

Research Experience

University of Utah, Kahlert School of Computing, Salt Lake City, UT

Graduate Research Assistant, August 2023 – Present
Advisor: [El Kindi Rezig](#) (Postdoc, MIT CSAIL; Ph.D., Purdue University)

- Developed algorithms for qualitative join discovery in data lakes using example-based queries
- Built systems for semantic table sorting, natural-language intent-aware join discovery, and location-aware join discovery over data lakes
- Implemented scalable data wrangling systems with direct manipulation interfaces, handling data transformations

Bangladesh University of Engineering and Technology, CSE, Dhaka, Bangladesh

Research Assistant, July 2022 – June 2023
Advisor: [Muhammad Abdullah Adnan](#) (Ph.D., University of California San Diego)

- Developed machine learning pipelines for processing and analyzing video datasets for sign language translation
- Built data collection and annotation systems for creating structured datasets, handling data cleaning

Technical Skills

Data Systems & Databases: PostgreSQL, MySQL, MongoDB **Data Processing & ML:** PyTorch, TensorFlow

Cloud & Infrastructure: Docker, Google Cloud Platform

Programming Languages: Python, JavaScript/TypeScript, C++, SQL

Development Tools: Node.js, Express.js, React.js, Vue.js, Streamlit

Additional Experience

Everforth, Tokyo, Japan (Remote)

Frontend Developer, April 2022 – June 2023

- Built e-commerce web applications using Vue.js and CakePHP